Extra credit problems

Math 471

0. Find a mistake or misprint in the book. (The score depends on the type of mistake.)

1. Show that a composition of three reflections in the sides of a nondegenerate triangle does not have a fixed point.

2. Given disticnt points A, B and X make a ruller-and-compass construction of another point $Y \in (AX)$ such that $AY + BY = AX + BX$ and explain why it works. (You may play with the GeoGebra applet apfzzybw.)

3. Let m be an indirect motion of the plane. Show that the midpoints of all line segments $[X\, m(X)]$ lie on one line.

4. Let us denote by ρ_ℓ the reflection across line ℓ. Let ℓ and m be lines on the plane and n is the reflection of ℓ across m. Show that $\rho_m \circ \rho_\ell = \rho_n \circ \rho_m$.

5. Lines ℓ and m are tangent to two circles of radiuses r and R on such a way the circles are on one side of ℓ and on different sides of m. Let A and B be tangential points of ℓ and Q be the point of intersection ℓ and m. Show that $QA \cdot QB = R \cdot r$.

6. Given a line segment with marked midpoint, perform a ruler-only construction a line through a given point parallel to the the segment. Explain why it works. (You may play with the GeoGebra applet qrnffpkv.)

7. Let ABC be a nondegenerate triangele and $A' \in (BC)$, $B' \in (CA)$, $C' \in$ $\in (AB)$ be the points such that

$$2 \cdot \measuredangle AA'B \equiv 2 \cdot \measuredangle BB'C \equiv 2 \cdot \measuredangle CC'A \equiv \tfrac{2}{3} \cdot \pi.$$

Show that the triangle formed by the lines (BB'), (CC'), and (AA'), is congruent to $\triangle ABC$.

8. Give a ruler-and-compass construction of an inscribed quadrilateral with given sides. (You may play with the GeoGebra applet q9xkwnwb.)

9. A circle Γ with marked center O is given. Let ℓ be a line that passes thru O, and P be a point on Γ. Make a ruler-only construction of a line thru P that is perpendicular to ℓ. (You may play with the GeoGebra applet ujaxcm44.)

10. Let Γ be a circle with the center O and let A and C be two different points on Γ. For any third point P of the circle let X and Y be the midpoints of the segments AP and CP. Finally, let H be the orthocenter of the triangle OXY. Prove that the position of the point H does not depend on the choice of P.

11. Two points A and B lie on one side of a line ℓ. Two points M and N are chosen on ℓ such that $AM + BM$ is minimal and $AN = BN$. Show that points A, B, M and N lie on one circle.

12. Suppose D and E lie on the same side from (AC), $(AE) \parallel (CD)$, and $AB = BC$. Let K be the intersection of the bisectors of the angles EAB and BCD. Prove that $(BK) \parallel (AE)$.

13. Show that a neutral plane is Euclidean if and only if it has a rectangle.